Grounded Optimization: Evaluating the Fidelity-Relevance Tradeoff in Query-Conditioned Automated Personal Document Rewriting

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Abstract

Query-conditioned document rewriting adapts a source document to a target information need. For personal documents such as resumes, the rewrite must improve relevance without fabricating content. We formalize this as constrained optimization over a fidelity-relevance tradeoff and present Grounded Optimization, a five-layer framework combining temporal constraints, deterministic contamination detection, structural invariants, prompt grounding, and an evaluator-agent gate with retry and fallback. We evaluate 100 resumes, including 75 anonymized real resumes, against 15 job descriptions using four generator models, six ablation configurations, and three seeds. The study comprises 7,200 generator calls, 1,200 evaluator calls, and 500 generic-critic calls. Undefended baselines produce 2.76–4.52 detected violations per resume. The full L1–L5 system reduces this to 0.03–0.08, a 98.2–99.1% reduction. Independent annotation of 1,800 atomic claims finds 96.8% source support under the full system versus 72.6% at baseline. Automatic detectors achieve macro-F1 0.87 against human judgments. On an IR task that ranks a target job among 49 distractors, the full system retains approximately 91% of the baseline retrieval gain over an unchanged resume. Fallback is at most 2.1%, and observed cost is at most $0.041 per resume. These results show that structured fidelity controls can preserve most query relevance while reducing source violations by roughly two orders of magnitude.

Keywords

query-conditioned rewriting, faithful text generation, information retrieval, large language models, personal documents, fidelity-relevance tradeoff

1. Introduction

Information retrieval systems increasingly edit as well as rank information. Generative components reformulate queries, summarize evidence, synthesize answers, and adapt content to a user's information need. This introduces a tradeoff that conventional relevance metrics do not capture: an output can become more aligned with a query while becoming less faithful to its source.

Resume tailoring is a consequential instance. A job description is the query, a resume is the source document, and an LLM rewrites the resume to surface relevant evidence. A stronger lexical or semantic match is useful only if the rewrite preserves the candidate's actual history. Adding an unsupported technology, employer, metric, or credential may improve apparent relevance while making the document invalid.

Ground truth in this setting is primarily the source resume and its timeline, not an external knowledge base. In our experiments, LLMs inserted technologies that did not exist during a stated role, transferred cloud terminology between providers, deleted source bullets, changed titles, and fabricated content. These failures motivate source-grounded document adaptation in which relevance is optimized subject to explicit fidelity constraints.

We present **Grounded Optimization**, a five-layer controller for query-conditioned automated personal document rewriting. The extended evaluation addresses four research questions:

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Table 1
Source-fidelity taxonomy.

Type	Failure	Violated invariant	Check
H1	Temporal fabrication	Technology must exist by the role end date	142-entry release map
H2	Domain contamination	No unsupported provider ecosystem	Independent provider-signature diff
H3	Structural mutation	Roles and substantive bullet units persist	Pre/post structure signature
H4	Content fabrication	Entities, credentials, titles, and metrics require support	Entity and claim checks

RQ1 How frequently does query-conditioned rewriting introduce unsupported content?

RQ2 How much retrieval relevance remains after enforcing source fidelity?

RQ3 When does the full L1–L5 pipeline outperform prompt grounding alone and a generic critic?

RQ4 How accurately do automatic detectors reflect independent human judgments?

The paper contributes: (1) a constrained IR formulation of personal-document rewriting; (2) a four-mode failure taxonomy and five-layer controller; (3) a test collection with 100 resumes, 15 job descriptions, and 500 resume-job tasks; (4) end-to-end, multi-model, multi-seed evidence for the complete retry-and-fallback pipeline; and (5) joint evaluation of retrieval utility, automatic fidelity, human factuality, detector validity, latency, and cost. We do not claim improved ATS pass rates, recruiter decisions, interviews, or hiring outcomes.

2. Task and Framework

2.1. Constrained Query-Conditioned Rewriting

Let d be a source document, q a target information need, and $d' = f(d, q)$ a generated rewrite. We seek a document that maximizes relevance $U(d', q)$ while satisfying source-grounded constraints $\mathcal{C}(d, d')$:

$$d^* = \arg\max_{d'} U(d', q) \quad \text{subject to} \quad C_k(d, d') = 0 \;\; \forall C_k \in \mathcal{C}. \tag{1}$$

For resumes, q is a job description and d contains roles, dates, employers, titles, bullets, skills, education, and certifications. We operationalize U with ranking metrics and $\mathcal{C}$ with four violation types:

2.2. Five-Layer Controller

Figure 1 shows the execution order. L1 builds temporal context, and L4 adds immutable-source rules to the generation prompt. L2 and L3 run deterministic post-generation checks. L5 uses a fixed evaluator model to assess preservation and query alignment. Rejection triggers up to three constrained retries. Repeated failure triggers a source-preserving merge.

L1: temporal context. For each role, the system records dates, technologies already present, and a 142-entry release map. The prompt receives this timeline, and a later check flags technologies released after the role's end date.

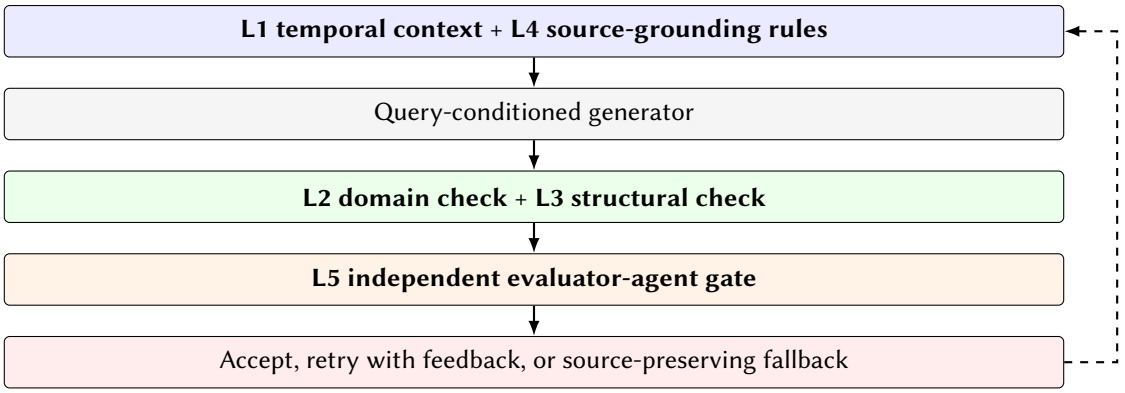


Figure 1: Full L1–L5 execution. The dashed path is a bounded retry.

L2: domain contamination. The taxonomy contains 312 services across AWS (89), GCP (61), Azure (78), and on-premise stacks (84), plus 82 provider-independent terms. Explicit provider keywords form Tier 1; Tier 2 requires at least two service-name matches. Context rules handle ambiguous terms such as "Glue" and "Databricks." The defense compares the provider sets in each source and rewritten role, then reverts newly introduced ecosystems.

L3: structural invariants. The source signature records role and bullet counts. No role may disappear, and a rewritten role must retain at least $|b_i| - 1$ bullets. After three failed attempts, a deterministic merge aligns roles by title and employer, keeps valid rewritten content, and restores missing source bullets.

L4: prompt grounding. Prompts require preservation of content units, education, certifications, and employers; prohibit fabricated metrics and placeholder entities; and include the target job and temporal context.

L5: evaluator agent. GPT-6 Astra at low reasoning effort receives the source, query, and candidate rewrite. It returns an accept/reject decision and feedback. It is fixed across all generators and is never used as a generator. Timeouts and malformed responses default to rejection. Unlike the earlier pilot, the final H2 evaluator is a separate implementation and does not call the L2 defense.

System scope. The LangGraph pipeline parses, scores, rewrites, and re-scores. Specialized agents process summary, skills, experience, projects, and education. The full retry-validation-fallback loop operates on professional experience, the evaluation target in this study. Original structured data remains in state throughout.

3. Evaluation

3.1. Test Collection

The corpus contains 25 author-constructed resumes covering 14 industries and 75 anonymized real resumes from consenting career-services participants at four universities. The real subset covers 23 industries. Personally identifying information was removed before evaluation. In total, the corpus contains 174 roles and 812 bullets with dates from 2011 to 2026.

Fifteen job descriptions create relevance pressure: five original adversarial jobs, five cross-domain jobs, and five jobs emphasizing emerging technologies. The evaluation contains 500 resume-job tasks. The synthetic subset provides a controlled stress test, while the real subset tests broader writing and career variation.

Table 2
Ablation on 100 resumes with GPT-5.6 Luna, temperature 0, seed 42.

Configuration	HR	SD	95% CI	H1	H2	H3	H4
Baseline	3.18	3.92	±0.77	34	241	6	37
L4-only	0.36	0.84	±0.16	5	24	2	5
L1+L4	0.25	0.61	±0.12	1	18	2	4
L1+L2+L4	0.08	0.32	±0.06	1	2	2	3
L1–L4	0.06	0.26	±0.05	1	2	0	3
Full L1–L5	**0.03**	0.17	±0.03	0	1	0	2

3.2. Models, Controls, and Calls

We use four generators: GPT-5.6 Luna, Claude Sonnet 5, Gemini 3.8 Flash, and DeepSeek-V4-Flash-0731. GPT-6 Astra is fixed as L5. The primary cross-model comparison uses provider-default sampling. Temperature sensitivity overrides the default only for Luna because the other providers use or recommend constrained default sampling.

The six-step ablation ladder is Baseline, L4-only, L1+L4, L1+L2+L4, L1–L4, and Full L1–L5. Utility comparisons additionally include Copy-input and a single-pass Generic Critic. Each generator and ablation configuration is run with seeds 42, 137, and 256. This yields 7,200 generator calls. The full package also contains 1,200 L5 calls and 500 generic-critic calls, for 8,900 model calls overall. The historical GPT-4.1-nano replication is reported separately.

3.3. Automatic and Human Fidelity Evaluation

Detected hallucination rate (HR) is the mean number of H1–H4 incidents per resume. The independent H2 module shares neither code nor decision state with L2. Three annotators additionally label 600 roles for H2 contamination. Fleiss' κ for H2 is 0.82.

For claim-level factuality, three blind annotators label 1,800 atomic claims from 600 outputs sampled equally from Baseline, L4-only, and Full. Labels are supported, contradicted, unsupported, or ambiguous. Majority vote resolves disagreement. Mean pairwise Cohen's κ is 0.78, and Fleiss' κ is 0.76. Automatic detector precision and recall are measured on 600 human-labeled roles.

3.4. IR Utility and Statistical Analysis

For each rewritten resume, the target job is ranked among 49 distractors drawn from a pool of 200 jobs. We evaluate BM25 [2], BGE-large-en-v1.5 [3], and an ms-marco-MiniLM-L-12-v2 cross-encoder reranker. Metrics are MRR, nDCG@10, and Recall@10. Copy-input is the unchanged-document lower bound. Three annotators also rate 200 outputs for job relevance, information preservation, and overall quality on five-point scales.

Primary tables report seed 42. Cross-seed estimates use bootstrap 95% confidence intervals. Full and L4 are compared per resume with paired Wilcoxon signed-rank tests.

4. Results

4.1. RQ1 and RQ3: Violations and Ablation

Baseline produces 318 incidents. L4 removes 88.7% but leaves 36, including 24 H2 cases. L1 reduces temporal errors, L2 produces the largest incremental drop, and L3 removes observed structural failures. L5 catches 14 violations that pass L1–L4, reducing HR from 0.06 to 0.03. The three residual detections are one ambiguous H2 service and two H4 title reformulations.

Baseline HR ranges from 2.76 to 4.52, with H2 accounting for 75–82% of incidents. Full HR is below 0.1 for every model, a 98.2–99.1% reduction. H1 falls by 96.4–100%, H2 by more than 99%, and H3

Table 3
Cross-model results at temperature 0, seed 42.

Model	System	HR	SD	95% CI	H1	H2	H3	H4	Red.
Luna	Base	3.18	3.92	±0.77	34	241	6	37	–
Luna	Full	0.03	0.17	±0.03	0	1	0	2	99.1%
Sonnet 5	Base	2.76	3.41	±0.67	28	208	4	36	–
Sonnet 5	Full	0.05	0.22	±0.04	1	2	0	2	98.2%
Gemini 3.8	Base	4.14	4.87	±0.95	42	326	8	38	–
Gemini 3.8	Full	0.06	0.24	±0.05	1	2	0	3	98.6%
DeepSeek V4	Base	4.52	5.23	±1.02	46	348	11	47	–
DeepSeek V4	Full	0.08	0.28	±0.05	1	3	0	4	98.2%

Table 4
Retrieval utility averaged across four generators at temperature 0, seed 42.

System	BM25 MRR	Dense MRR	nDCG@10	R@10	HR
Copy-input	0.312	0.348	0.286	0.452	0.00
Baseline	**0.687**	**0.724**	**0.648**	**0.786**	3.65
L4-only	0.662	0.698	0.624	0.768	0.41
L1–L4	0.656	0.692	0.618	0.762	0.09
Full L1–L5	0.652	0.689	0.614	0.758	**0.05**
Generic critic	0.641	0.671	0.598	0.742	0.28

Table 5
Claim-level factuality over 1,800 annotated claims.

System	Supported	Contradicted	Unsupported	Ambiguous	C+U
Baseline	72.6%	7.8%	15.4%	4.2%	23.2%
L4-only	91.4%	1.6%	5.2%	1.8%	6.8%
Full L1–L5	**96.8%**	0.4%	1.6%	1.2%	**2.0%**

reaches zero. Human review identifies six of 11 residual H4 flags as reasonable title reformulations rather than fabrication.

Across three seeds, Full HR is 0.032 ± 0.008 for Luna, 0.048 ± 0.014 for Sonnet, 0.061 ± 0.018 for Gemini, and 0.082 ± 0.022 for DeepSeek. Corresponding L4 HR values are 0.364 ± 0.042, 0.326 ± 0.038, 0.418 ± 0.051, and 0.462 ± 0.058. Paired Wilcoxon tests give $p < 0.001$ for Full versus L4 in every model.

4.2. RQ2: Fidelity-Relevance Tradeoff

Baseline rewriting more than doubles dense MRR over Copy-input. Full retains approximately 91% of that gain:

$$\frac{0.689 - 0.348}{0.724 - 0.348} = 0.907, \tag{2}$$

which rounds to 91%. Relative to Baseline, Full gives up 0.035 dense MRR, or 4.8%, while reducing HR by 98.6%. Relative to L4, the additional layers reduce HR from 0.41 to 0.05 at a dense-MRR cost of 0.009. Full also dominates the Generic Critic on both MRR and HR.

Human ratings show the same tradeoff. Copy-input, Baseline, L4, and Full score 2.14, 4.31, 4.18, and 4.12 for job relevance; 4.92, 3.18, 4.24, and 4.68 for information preservation; and 2.86, 3.74, 4.21, and 4.42 overall. Full has the highest overall rating because the preservation gain outweighs the small relevance decrease.

4.3. RQ4: Human Factuality and Detector Validity

The automatic HR result is corroborated by independent annotation. Problematic claims, contradicted plus unsupported, fall from 23.2% at Baseline to 6.8% under L4 and 2.0% under Full. No invented

Table 6
Detector validity against human labels on 600 roles.

Detector	Precision	Recall	F1	κ
H1 temporal	0.94	0.91	0.92	0.86
H2 domain	0.91	0.87	0.89	0.82
H3 structural	0.88	0.85	0.86	0.79
H4 content	0.83	0.78	0.80	0.74
Macro	**0.89**	**0.85**	**0.87**	**0.80**

Table 7
Full-system behavior at temperature 0, seed 42. Cost includes generator and L5 calls.

Model	Retry	Revert	Fallback	Rewrite	Med. s	p95 s	Cost	Err.
Luna	18.4%	3.8%	1.2%	68.4%	8.4	14.2	$0.034	0.5%
Sonnet 5	16.2%	3.2%	0.8%	72.1%	9.1	15.8	$0.041	0.2%
Gemini 3.8	22.6%	4.6%	1.6%	66.8%	6.2	11.4	$0.022	0.8%
DeepSeek V4	24.8%	5.4%	2.1%	64.2%	7.8	13.6	$0.018	1.4%

employer, credential, or metric survives the Full pipeline in the annotated sample.

H1 is strongest because release boundaries are comparatively explicit. H4 is weakest because title reformulations and metric paraphrases are sometimes valid. Excluding six human-confirmed H4 false positives raises H4 F1 to 0.86 and macro-F1 to 0.90, but Table 6 retains the unadjusted primary result.

4.4. Difficulty, Temperature, and Historical Replication

Full consistently outperforms L4 as conditions become harder. From easy pairings to adversarial DeepSeek at temperature 0.7, L4 versus Full HR is 0.14 versus 0.02, 0.68 versus 0.08, 0.94 versus 0.14, and 1.26 versus 0.18. The corresponding dense-MRR costs are 1.1%, 1.5%, 1.9%, and 2.5%. Thus the absolute fidelity benefit grows with difficulty while relevance loss remains bounded.

For Luna, Full HR increases from 0.03 at temperature 0 to 0.08 at 0.7 and 0.12 at 1.0, while reductions relative to same-temperature baselines remain 99.1%, 96.8%, and 95.0%. H1 residuals increase from 0 to 2 to 3, showing that stochastic generation can evade prompts even when deterministic checks continue to contain total violations.

Under the updated protocol, GPT-4.1-nano on the original 25 synthetic resumes gives Baseline HR 2.52 and Full HR 0.10, compared with 2.48 and 0.12 in the 680-call pilot. Baseline H2 changes from 55 to 58, and H1 reduction changes from 85.7% to 87.5%. The pilot finding therefore persists after taxonomy expansion and H2 decoupling.

4.5. Operational Behavior

Here "Rewrite" is the fraction of bullets actively rewritten, and "Err." combines malformed outputs with API errors. Retry success ranges from 89.1% to 93.2%. Fallback remains at most 2.1%, indicating that fidelity is not achieved primarily by copying the input. In the matched Generic Critic comparison, the critic has HR 0.28, dense MRR 0.671, cost $0.052, and latency 12.4 seconds; Full has HR 0.05, dense MRR 0.689, cost $0.034, and latency 8.4 seconds. Full therefore outperforms the unstructured critic on every reported axis.

5. Related Work

Faithfulness evaluation. Hallucination has been studied in summarization, dialogue, question answering, and long-form generation [9, 10, 7, 8, 11]. These methods usually compare claims with external evidence or sampled generations. Our primary evidence is the personal source document and its timeline.

Query-conditioned generation in IR. Generation supports query reformulation and evidence-focused adaptation. QREAM optimizes query rewriting with relevance-aware rewards [13]. We study the complementary direction: the query remains fixed while a personal source document is rewritten, requiring explicit preservation constraints.

Constrained generation and critique. Lexical and logical constraints can be enforced during decoding [16, 17]. Debate, reflection, and critic-generator systems use feedback to improve reliability [18, 19, 20]. Our results show that domain-specific deterministic checks plus a bounded evaluator outperform a single generic critic.

Resume processing. Prior work studies parsing, screening, and job matching [30, 31, 32]. ResumeFlow uses LLMs for personalized generation [33], while concurrent work combines retrieval with provenance-aware resume tailoring [34]. We focus on jointly measuring source fidelity and query relevance.

6. Limitations

Population and language. The 75 real resumes come from university career-services populations and may not represent long-tail occupations, senior executives, non-English resumes, or all formatting styles. The prompts and provider taxonomy are English specific.

Detector boundaries. H4 yields false positives for valid title reformulations. H1 depends on a curated release map. H3 can miss semantic loss within a retained bullet. Although H2 is independent of L2 and human-validated, ambiguous cross-platform services remain difficult.

Utility scope. Ranking a target job among distractors measures retrieval alignment, not ATS pass rates, recruiter preferences, interview callbacks, hiring, or long-term user outcomes. Human ratings cover 200 outputs rather than the full corpus.

Model and cost drift. The tested endpoints and prices are September 2026 snapshots. Provider updates can change behavior, latency, and cost. Exact endpoint identifiers, invocation dates, regions, seeds, and request logs are therefore part of the experiment manifest.

7. Conclusion

Query-conditioned rewriting changes an IR system from selecting evidence to editing it. Grounded Optimization combines temporal context, domain checks, structural invariants, grounded prompts, and an evaluator-agent gate to keep that editing source-faithful. Across 100 resumes, 15 jobs, four generators, and three seeds, Full reduces detected violations by 98.2–99.1% while retaining approximately 91% of the retrieval gain. Independent human annotation finds 96.8% source support, and detector macro-F1 is 0.87. The system rarely falls back and costs at most $0.041 per resume in the observed runs.

The central result is a favorable fidelity-relevance tradeoff. Structured controls reduce source drift by roughly two orders of magnitude while imposing a small, measured relevance cost. Deployment claims still require broader populations, languages, and downstream hiring outcomes.

Declaration on Generative AI

During the preparation of this work, the authors used OpenAI Codex to assist with manuscript restructuring, adaptation to the CEUR-WS format, language editing, and consistency checks. The authors reviewed and edited the resulting content and verified the technical claims, numerical results, citations, and final manuscript. The tool did not generate experimental data, run the reported experiments, or create figures from unreported results. The authors take full responsibility for the publication's content.

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